Final Year Project
Project Title Registration Form

Student name:	John
Student surname:	Gambino
Student number:	14002996
Programme of Studies:	BSc Computing with Foundation Year (full-time)
Project Title:	Design and Evaluation of an Adaptive Multi-Model Orchestration System for Heterogeneous LLM Workloads

Project introduction and description:

This project will focus on designing and evaluating an adaptive multi-model orchestration system. When a user or software agent submits a task to the system, the orchestration layer will determine how that task should be routed between different local and cloud-based LLMs. The aim of the proposed system is to identify which model, or strategy for execution, is best suited for any given task, while ensuring that response quality, raw usage cost and the time taken for a response to be given are balanced and appropriate for the task. In the scenario that a highly complex task is provided to the system, it may choose to decompose the task into smaller subtasks for routing or escalate the task to a higher-level LLM if the initially chosen model is not suitable for the task. The system will be tested against a series of standardised baselines, including comparison to a single-model execution approach, whilst tracking and logging system performance, usage costs and response times to be used for project evaluation. In essence, this project will assess whether the usage of multiple LLMs when executing tasks can provide more efficient responses to the user while not resulting in their degradation. The author of this project is the right person to do so, due to previous experience with software development, and experimentation with local and cloud-based AI models.